

Characterisation of powder bulk, dynamic flow and shear properties in relation to die filling

R. Freeman and Xiaowei Fu

Understanding powder behaviour during preparation, transfer and filling of dies is important in powder metallurgy since each step influences the quality of the final component. For high productivity and quality during manufacture, it is essential that process and powder characteristics are compatible. This investigation of the bulk, dynamic flow and shear properties of four materials demonstrates that die filling effectiveness correlates well with many powder properties. Thus the availability of accurate information about the flow and bulk properties of powders should allow reliable prediction of die filling efficiency, removing any need for the more usual 'trial and error' approach.

Components made from metal powders are manufactured in distinct stages:

- die filling – powder drops from a feed shoe into the die cavity
- powder transfer – powder is transferred within the die and through a series of tool motions to produce a compact approaching the final shape
- compaction – powder in the die is compressed to form a green body
- ejection – the green body is ejected from the die
- sintering – the green body is sintered, often in a reducing atmosphere
- secondary treatments or finishing – the component is sized or machined to ensure the required dimensional tolerance is achieved.

The compaction, ejection and sintering stages in particular have a significant impact on the properties of the finished components. A knowledge of the behaviour of powders in preparation, die filling and powder transfer is

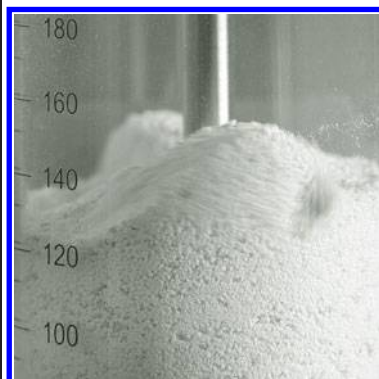
important, since their packing structure and density distribution may influence later process steps, and affect the integrity of the final components.

Various experimental techniques have been used to measure powder flowability in relation to die filling. These include: Hausner ratio and Carr index (poured and tapped bulk density);

Hall or Flodex flowmeters (mass flowrate or time required to discharge through an orifice); angle of repose; and the shear cell, which measures the yield strength of a consolidated bulk solid. While all may be useful in specific process environments, none predicts the behaviour of a powder during die filling.

FT4 powder rheometer

The FT4 Powder Rheometer provides comprehensive powder flowability data, delivering automated shear testing, dynamic flowability and bulk properties measurements.



FT4 works on patented blade principle: twisted blade displaces powder as it moves along a helical path through sample

Dynamic testing

The FT4 works on a patented blade principle, a twisted blade displacing powder as it moves along a helical path through the sample. Depending on the direction and speed of movement, a broad range of flow patterns and rates can be achieved. Axial and rotational forces acting on the blade are measured and these data form the basis of the flowability assessment. Force measurements are converted into energy to determine the total energy consumed or work done during the traverse.

Shear testing

Shear testing has traditionally been

used to determine the behaviour of consolidated powders. The shear properties measured, which include cohesion values, compression strength (unconfined yield strength) and internal angle of friction, are all useful powder characteristics that complement dynamic flow energy measurements and bulk properties data.

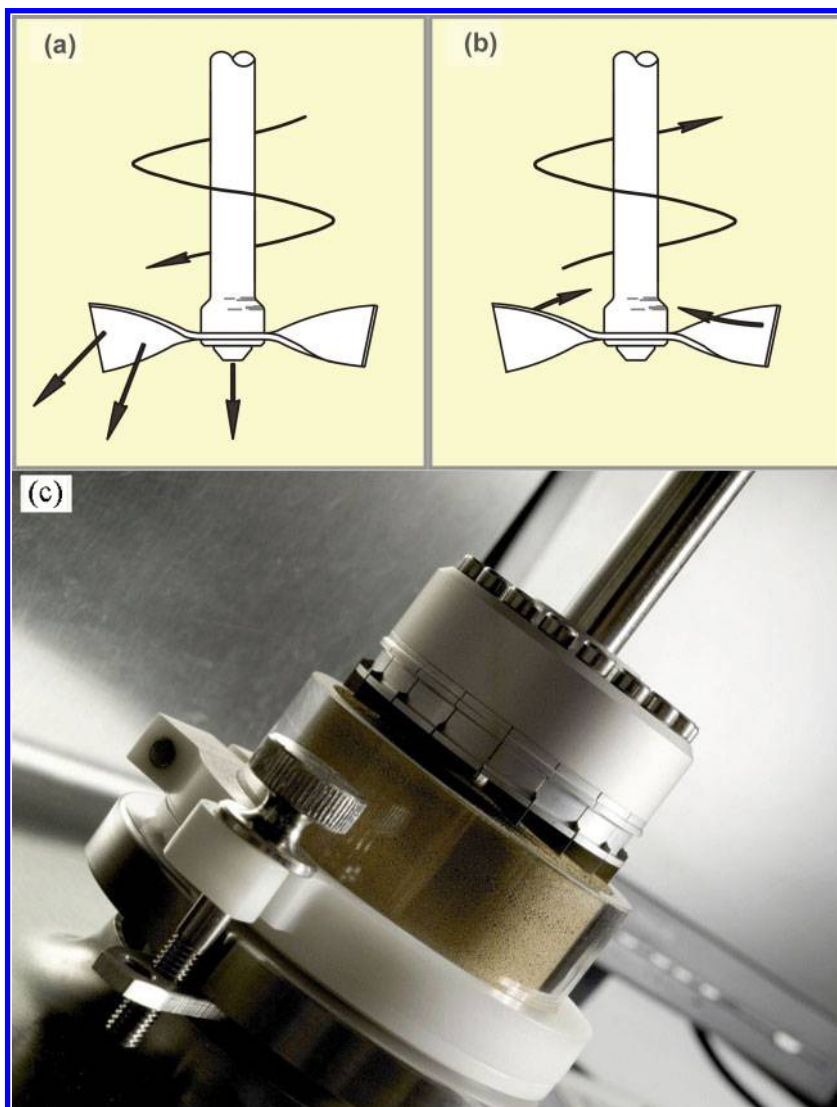
Bulk properties

Automated testing enables the precise measurement of bulk density, permeability and compressibility. Bulk properties characterisation includes: bulk density measurements on conditioned, tapped and consolidated powders; permeability (pressure drop at fixed air throughput) as a function of varying bulk pressure; and compressibility of a powder bulk as a function of external pressure.

Sample conditioning

Regardless of which type of measurement is being carried out – dynamic, shear or bulk property determination – a prerequisite is a robust methodology that ensures the material is always analysed in the same state. Sample conditioning is a core element of every automated testing procedure. The conditioning step is a downward and upward traverse of the blade, which loosens and slightly aerates a compacted sample or releases air from overly aerated powders, ensuring that all measurements are carried out using a precisely known volume of homogeneously packed powder.

Reg Freeman is Managing Director and Xiaowei Fu Research Scientist at Freeman Technology Ltd, Boulsters Farm Centre, Castlemorton Common, Welland, Worcs. WR13 6LE, UK, tel. +44 (0) 1684 310860, fax +44 (0) 1684 310236, Internet www.freemantech.co.uk



1 Methodology of powder flow testing with FT4 powder rheometer: a downwards testing mode showing bulldozing action along the entire blade length; b upwards testing – shearing with minimal consolidation; c shear cell above sample vessel

Powder behaviour is complex. It is influenced by a combination of physical properties and the characteristics of the processing equipment, and powder flowability cannot be expressed adequately as a single value or index. Die filling is a dynamic process, so any powder characterisation methods used should closely reflect the real industrial situation.

In the study described below, the die filling behaviour (die filling ratio) of tungsten, aluminium, and two kinds of glass beads of different nominal size

was examined. Bulk, dynamic flow and shear properties were characterised using a FT4 powder rheometer (Freeman Technology).

Experimental approach

Particle size and morphology

Particle size distribution for each material was determined using a Mastersizer 2000 laser diffraction instrument (Malvern Instruments, UK). Particle morphology was characterised with a JEOL 6340F scanning electron

microscope (SEM). The results are given in Table 1.

Powder conditioning

Obtaining repeatable data requires powders to be in a homogeneous packing state. In the as supplied state, a powder has a unique history that has been influenced by factors such as consolidation, segregation, aeration, humidity, caking, or vibration. The 'conditioning' needed to remove this history involves gentle displacement of the whole powder sample, loosening and slightly aerating it into a homogenised and reproducible state.

All samples for dynamic, bulk and shear testing were conditioned and 'normalised' before measurement, using the FT4 powder rheometer.

Flowability measurement

The FT4 is a universal powder tester (see box). Accessories such as blades, pistons and shear heads can be rotated and simultaneously moved axially into a powder sample while axial and rotational force are measured. A number of control modes are available on both axes, including velocity, force and torque. Standard dynamic tests, aeration testing and shear testing are automated, with no operator involvement apart from sample preparation.

Dynamic testing was performed using a 48 mm diameter blade and a 160 mL powder sample contained in a 50 mm bore borosilicate test vessel (Fig. 1a and b). An automated, 18 segment, 48 mm diameter rotational shear cell accessory (Fig. 1c) was used for all shear testing, using 85 mL samples.

Die filling rig

A model die filling rig was designed to mimic commercial die filling processes (Fig. 2). It consists of a stationary die and a motorised unit which drives the shoe at a steady velocity from 50 to 300 mm s⁻¹. In this study, the feed shoe was cylindrical with a fixed volume of 160 mL and diameter of 50 mm. The cylinder die had a 10 mL volume and 25 mm diameter.

Experiments were carried out in air at shoe velocities of 50–250 mm s⁻¹. The 160 mL samples were conditioned on the FT4 then carefully transferred to the rig for translation over the die. The mass transferred into the die was measured to determine the filling ratio. This was repeated three times, each time using a newly conditioned sample of powder.

Table 1 General powder properties

Material/powder description	D_{50} , μm	Shape
GL glass beads	174	Spherical
GS glass beads	68	Spherical
Aluminium powder granules	134	Irregular
Tungsten powder	4	Angular

Results and discussion

Powder flowability

The most important material properties affecting powder flowability, summarised in Table 2, are further discussed in this section.

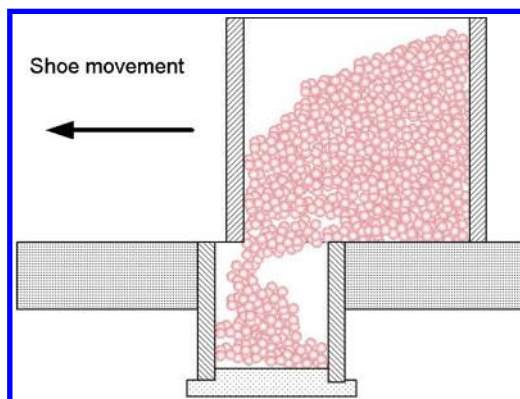
Specific energy

Specific energy (SE) is a measure of how easily a powder will flow in an unconfined or low stress environment. It is calculated from the energy needed to establish a particular flow pattern in a conditioned, precise volume of powder. During measurement, this flow pattern is an upward clockwise motion of the blade (see Fig. 1b), generating gentle lifting and low stress flow. The SE is calculated from the work done in moving the blade through the powder from bottom to top of the vessel (upward traverse). Gravity dominates in this test, so to compensate for varying bulk densities the flow energy is expressed as specific energy, normalised by the mass of powder.

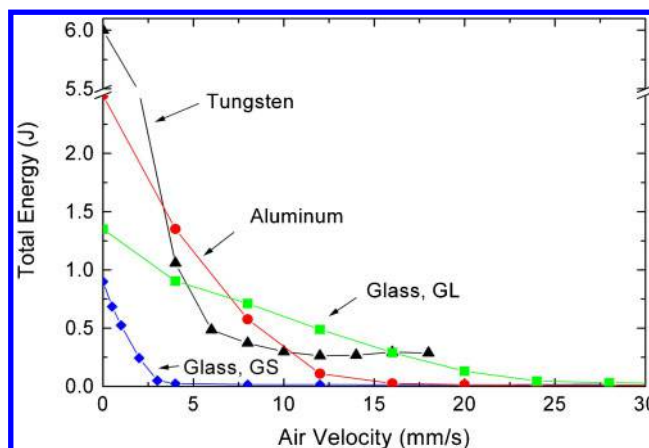
Specific energy depends primarily on the shear forces acting between particles. Cohesion is often the most influential property in low stress environments. Sample GS has the lowest SE, indicating it flows most readily in a low stress, conditioned state. Tungsten has the highest SE, suggesting it behaves most cohesively. Interestingly, GL has a higher SE than GS, indicating higher cohesion.

Basic flow energy (BFE)

In these conventional dynamic tests, a previously conditioned powder was consolidated using a bulldozing blade action (Fig. 1a) that forced the powder down towards the bottom of the



2 Schematic drawing of die filling process



3 Variation of flow energy as function of air velocity

containing vessel. The BFE is a key parameter that is highly sensitive and differentiating in relation to small differences in flow properties. Results varied from 5964 mJ (tungsten) to 899 mJ (Glass GS) as shown in Table 2.

Aeration

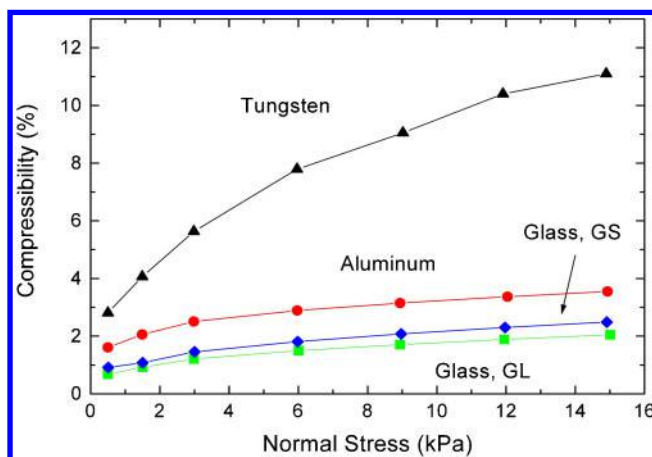
The presence/absence of air greatly affects a powder's flow properties; air is added naturally when powder is

moved freely. When a powder is aerated, less energy is normally required to move it. This reduction in flow energy is described by the aeration ratio (AR).

Cohesive powders do not allow air to permeate through the powder bulk – instead channels or rat-holes occur. The resulting change in energy is therefore small. In less cohesive powders air permeates the entire bulk, with a consequent large reduction in energy.

Table 2 Powder flow properties for investigated materials

	Glass GL	Glass GS	Aluminium	Tungsten
Basic flowability energy (BFE), mJ	1431	899	3300	5964
Stability index (SI)	1.03	0.97	1.25	1.10
Flow rate index (FRI)	1.04	1.02	1.48	1.40
Conditioned bulk density (CBD), g mL ⁻¹	1.44	1.46	1.24	4.17
Bulk density, consolidated by 20 taps, g mL ⁻¹	1.50	1.49	1.34	4.97
Specific energy (SE), mJ g ⁻¹	3.36	2.36	4.40	6.70
Aeration ratio (AR)	49.2	56.3	171.5	26.1
Pressure drop across powder bed at 2 mm s ⁻¹ air velocity (PD ₁₅), mbar	0.82	5.20	1.40	15.3
Consolidation index (CI _{20taps}) – factor by which flow energy increases relative to BFE	1.31	1.11	1.43	2.32
Volume change, 20 taps, %	4.0	2.0	7.5	16.1
Compressibility (volume change, 15 kPa direct pressure), %	2.03	2.46	3.50	11.1
Shear stress τ_2 , kPa	1.0	0.74	1.57	1.64
Shear stress τ_1 , kPa	0.52	0.39	0.88	1.02
Unconfined yield strength (UYS), kPa	0.28	0.13	0.45	1.09
Angle of internal friction, deg	23.9	18.7	35.7	34.0
Cohesion (Co), Pa	90	46	120	297
Flow function (FF _C)	14.0	29.3	11.6	5.53



4 Compressibility as function of applied normal stress

In some cases virtually all particles separate and the bed fluidises. The sensitivity of a powder to aeration relates well to its performance in a gravity feed system and for processes such as volumetric filling.

The four powders behave very differently during aeration when subjected to the same range of air velocities (see Fig. 3). Above a certain air velocity the aluminium, GL and GS glass beads fluidise. Because of its permeability, the GL sample requires the highest velocity, releasing air rapidly and needing considerable air flow to separate the particles.

The permeability of the smaller glass beads, GS, is lower by a factor of up to 6, allowing fluidisation at relatively low air velocity. Thus a small amount of entrained air will greatly improve the flowability of GS, compared with GL. However, GL will rapidly release a large amount of entrained air. The aluminium powder is also highly permeable, requiring high air velocity to fluidise. The tungsten with its poor permeability, does not fluidise, as shown by the high flow energy requirement (Fig. 3). Close bonding of the small tungsten particles makes the bed cohesive, with injected air forming channels to escape.

Effect of consolidation on flow energy

Closer packing and the loss of entrained air mean that compacted or consolidated powders are less likely to flow freely under gravity. Consolidation of all four powders was determined after subjecting each to 20 taps. Many processes (including die filling) impose vibration and the vulnerability of a powder to this is important. The standard dynamic test was used (as in the BFE measurement) to determine the extent to which the flow energy increased as a result of consolidation.

The results (Table 2) show an increase in normal flow energy (BFE) of 230% for the tungsten with a 16% volume reduction. This compares with an energy increase of 10% for GS glass and only a 2% volume reduction.

Compressibility

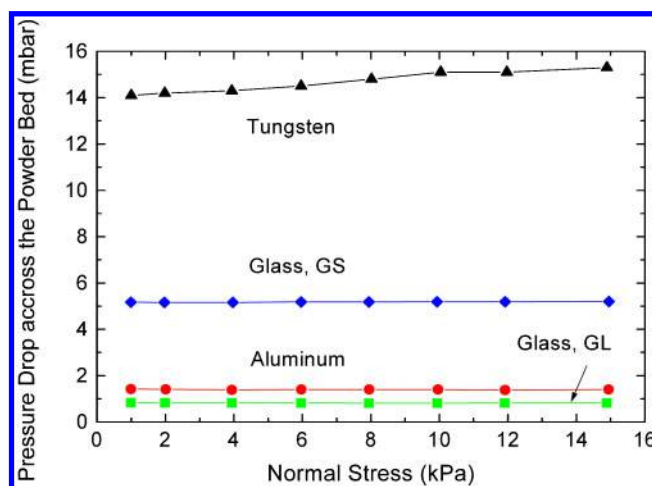
Compressibility is a bulk property measurement that examines the

volume change of a conditioned sample when it is slowly compressed so that entrained air can escape. While not a direct measure of flowability, it can indicate whether a powder is cohesive or free flowing.

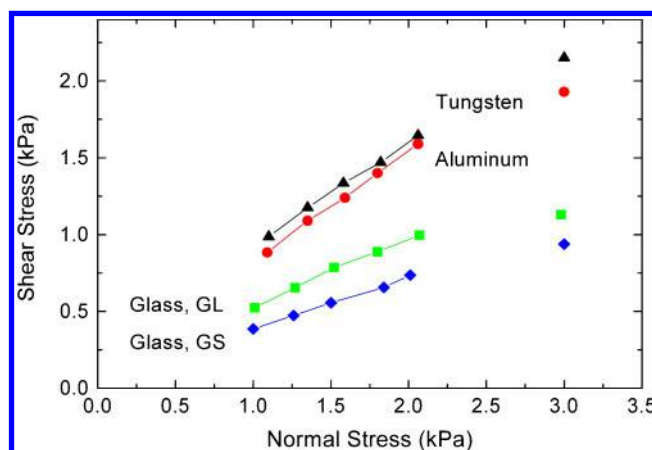
Tungsten is the most sensitive to compression (Fig. 4). GL and GS have very low compressibility, reflecting the relatively efficient packing of the particles when in a conditioned, low stress state. This is due to their spherical shape, low cohesion and high permeability. The aluminium powder has a median compressibility between glass beads and tungsten powder since its large particle size and irregular shape allow a degree of realignment and closer packing.

Permeability

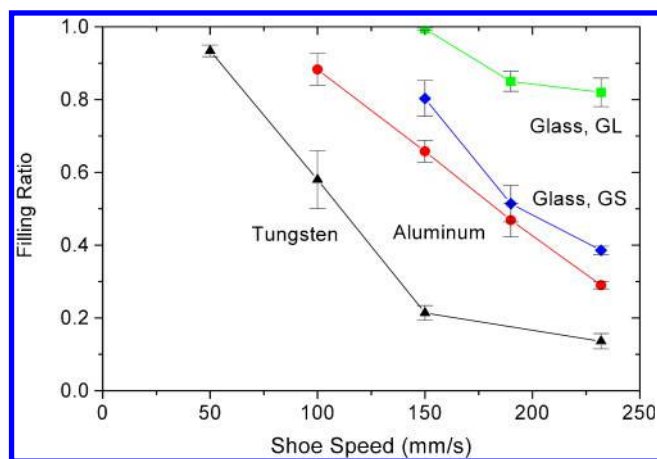
Permeability indicates the ease with which a material can transmit a fluid (in this case air) through its bulk. For powders, influencing factors include physical properties such as particle size and distribution, cohesivity, particle



5 Pressure drop through powder bed at constant 2 mm s^{-1} air velocity as function of applied normal stress



6 Shear stress versus normal stress of a sample consolidated and pre-sheared at 3 kPa normal stress



7 Die filling ratio of four different powders at different shoe speeds

stiffness, shape, surface texture and bulk density. External factors, consolidation stress for example, also have an effect by changing porosity and particle contact surface areas.

Generally, cohesive powders consisting mainly of sub-30 μm particles are the least permeable; granular powders are typically most permeable. Powders with large particles and fines may form a tight packing structure, with fines filling the spaces between particles, reducing the permeability of the powder bulk. Permeability determines the discharge rate of entrained air during die filling.

Figure 5 shows the air pressure difference required across the depth of the powder bed to maintain a constant air flow (2 mm s^{-1}) while consolidating the powder at increasing normal stresses. Increasing stress reduces the permeability of the tungsten but has little effect on the other powders due to their lower compressibility and greater particle size.

Granular aluminium and GL are more permeable than the other two samples

because of their relatively large void structure.

Shear properties

The primary flow mode of powders is shear, where particles slide relative to one another. High shear strength means greater resistance to flow. Figure 6 shows the relatively high yield loci of tungsten and aluminium compared with glass. Shear tests were carried out at 2, 1.75, 1.5, 1.25 and 1 kPa normal stresses, with samples pre-consolidated to 3 kPa normal stress. The derived unconfined yield strength or compression strength of the powders determines whether discharge from a hopper will cease due to the formation of a stable bridge. Cohesion value and internal angle of friction are also derivable from Fig. 6 (see Table 2).

Shear properties are important in die filling because powders retained in the dispensing shoe must shear under gravity in order to flow into the die, and must not bridge. These factors will determine the mass flow rate during

the die filling process. The higher the shear stress and internal angle of friction for a given normal stress, the more problematic the powder during flow initiation in the process environment.

Die filling behaviour

Conditioned powder

Gravity filling comprises two successive stages: powder flow from the shoe, and powder packing/settling in the die. Several major factors influence the die filling ratio, including powder flow properties, air escape from the die and escape of entrained air in the powder.

For the initial tests all powders were conditioned before die filling. The die filling ratios of conditioned powder were calculated from filled mass and conditioned bulk density (see Fig. 7). The four powders have very different die filling abilities: GL glass the best and tungsten the worst. At the lowest shoe speed even tungsten filled well, but at the highest speed the filling ratio deteriorated to below 20% compared with GL glass which remained above 80%.

Aerated and tapped powder

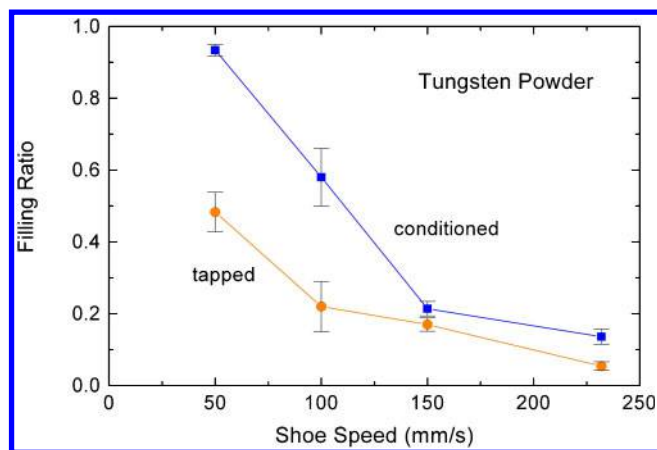
The packing condition of powder in the shoe and its impact on air content and bulk density affects die filling performance. To quantify this, aluminium and tungsten were evaluated in both slightly consolidated and lightly aerated states. Samples were aerated at 20 mm s^{-1} (aluminium) and 10 mm s^{-1} (tungsten) then transferred carefully to the die filling rig.

Aerated powder had the best filling ability and consolidated powder the worst.

The die filling ratio of tapped tungsten powder is shown in Fig. 8. The effect of tapping is dramatic, causing the die filling ratio to drop by a further 50%, and bulk density to increase by 16%. A single measurement on 'aerated' tungsten was made at 150 mm s^{-1} , showing no change from the conditioned data. Evidently aeration in tungsten powder produces channels and does not alter the packing state of the bed.

Powder properties and filling performance

Thirteen flowability and other powder properties were measured. For most parameters low values mean better flowability while higher mean poorer. The exception is aeration ratio where a



8 Die filling properties of tungsten powder under different powder packing conditions

high value indicates good aeration and flowability.

The availability of a diverse range of flowability indicators is particularly useful when comparing very similar materials. Because tungsten, aluminium and glass spheres are fundamentally different, it was not absolutely necessary here to measure all parameters. However the data (Table 2) show excellent correlation among all parameters – low values for the GS glass and the highest values for tungsten – confirming the poor flow properties of tungsten.

Figure 7 shows the filling ratios for the four powders when filling at 230 mm s^{-1} . The correlation is clear – tungsten has the worst flow properties and filling ratio, aluminium the next worst. The glass data are surprising: GS, with smaller spheres, has the best flow properties, yet inferior filling ratios, than the larger spheres of GL. This is attributed to the higher permeability of GL. Its ability to release entrained air instantly allows more complete filling of the die, confirmed by observing bed collapse rates.

Air is a major influence and particularly important in die filling by gravity. Crucial factors are:

- air content of the powder before filling
- air entrainment as powder falls from shoe to die
- release of the air entrained in the powder in the die
- escape of air from the die upon filling.

The first three relate directly to powder properties discussed above. Venting of the die volume depends on die design

and is not germane to the present study.

The importance of air on flow properties is demonstrated by the aeration data. Tungsten does not aerate due to its fine particle size and cohesivity. The glass spheres, however, are interestingly different, since GS aerates and eventually fluidises at relatively low levels of air addition, whereas GL, with higher permeability, requires a higher volume fraction of air to fluidise. GS therefore aerates rapidly on die filling, giving superior flow properties, but it is relatively poor at releasing air once in the die.

Aerating powder before filling confirmed that aeration improves flow properties (except for fine cohesive powders), improving flow into the die and giving higher filling ratios. Consolidation excludes air, increases bulk density, compromises flow properties and reduces the filling ratio.

Conclusion

The die filling performance of three very different materials has been measured and correlated with their flow properties and other characteristics. A comprehensive set of properties was measured using a universal powder tester, leading to the following conclusions.

There was clear correlation between die filling performance and powder properties. In general, powders having low values of flow energy, flowrate index, consolidation index, shear

strength, cohesion, compression strength, internal angle of friction and compressibility fill well. However, permeability is also important and there is an optimum permeability for powders that aerate, since too low a value will allow aeration but inhibit de-aeration. A highly permeable powder may not increase its entrained air or fluidise until a considerable air volume has been applied, but it will de-aerate rapidly.

The four powders have very different die filling performance; $174 \mu\text{m}$ GL glass is the best and $4 \mu\text{m}$ tungsten the worst. This is attributed to the good flow properties and excellent permeability of the GL glass beads. The cohesive $4 \mu\text{m}$ tungsten has poor flow properties and poor permeability.

Die filling performance depends on the packing condition of powders in the dispensing shoe. With $4 \mu\text{m}$ tungsten, consolidation significantly worsens flowability and may result in bridging and zero filling. Powders that readily entrain air will improve their flowability and die filling capacity if aerated.

In summary, die filling efficiency is predictable with a high level of confidence from information about both flow and bulk properties. Where powders that fill well are available alongside problematic powders, powder characteristics correlating with 'good' and 'bad' behaviour can be measured. New formulations may then be compared with this data and their die filling performance predicted.